INCH-POUND
MIL-DTL-24643/25F
1 October 2009
SUPERSEDING
MIL-DTL-24643/25E
22 August 2002

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -20 °C TO +150 °C, TYPE LSPI

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

Construction, Non-Watertight with Circuit Integrity

First	-	Copper conductor, nickel-coated, AWG 18, Class B stranding.
Second	-	Extruded silicone rubber insulation. Minimum average wall thickness 0.018 inch.
Third	-	Glass braid.
Fourth	-	Transparent braid covering.
Fifth	-	Two conductors (one colored black and one colored white for each pair of conductors) cabled together to form a pair with a lay not greater that 2.25 inches.
Sixth	-	An optional binder.
Seventh	-	Braided shield of 34 or 36 AWG uncoated copper over each pair. Minimum coverage of 85 percent.
Eighth	-	Shield insulation over each pair shield, consisting of polyethylene terephthalate film, 0.001 inch thick, applied helically with 50 percent overlap.
Ninth	-	A glass braid over each shield insulation (standard identification code applied by Method 4).
Tenth	-	The specified number of pairs (see table I) cabled together with a lay not greater than 14 inches. Fillers may be used as necessary to form a firm well-rounded assembly. Cabling sequence shall be consecutive, starting with pair 1 from the center outward.
Eleventh	-	A binder applied helically with overlap.
Twelfth	-	Silicone rubber jacket colored orange or red (see table I for thickness).
Thirteenth	-	Braided metal armor.

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TABLE I. Details.

Military part no. M24643/25	Type and size	No. of pairs	Overall cable	
			Jacket thickness min. avg. (inch)	Diameter max. (inch)
-01AN	LSPI-3	3	0.050	0.685
-02AN	LSPI-7	7	0.063	0.900
-03AN	LSPI-12	12	0.063	1.155

REQUIREMENTS:

Qualification required.

INSPECTION:

Basic Electricals:

Conductor resistance (ohms/1000 feet at 25 °C, max.)	7.5
Voltage withstand (volts, root mean square, min.)	
Conductor to conductor	1200
Conductor to shield	1200
Shield to shield	500
Insulation resistance (megohms/1000 feet, min.)	
Conductor to conductor	500
Conductor to shield	500
Shield to shield	100
Conductor continuity	No failure
Shield continuity	No failure

Group A:

Visual and dimensional	No failure

NOTE: The nickel coating on the conductors of the completed cable shall be permitted to have minor nicks, scrapes, or abrasions that are just perceptible to the eye.

Group B:

Cold bending cable	No failure
Thermoset proof test (percent, max.)	
Insulation	50
Jacket (when tested at 200 °C)	50
Physicals (unaged)	
Insulation (extruded)	
Tensile strength (lb/in^2, min.)	700

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Elongation (percent, min.)	150
Jacket (cable)	
Tensile strength (lb/in^2, min.)	800
Elongation (percent, min.)	250
Durometer (jacket) - Type A (hardness, min.)	60
Gas flame (1 hour)	No failure
Group C:	
Physicals (cable aging) (260±10 °C)	
Jacket (cable)	
Tensile strength (lb/in^2, min.)	600
Elongation (percent, min.)	150
Group D:	
Flame propagation (cable)	No failure
Qualification Inspection:	
Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:	
Gas flame (3 hours)	No failure
Aging and compatibility (cable) (125±5 °C)	No failure
Armor (conformance to material, construction and coverage)	No failure
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Insulation	0.2
Smoke index (max.)	
Jacket	35
Fillers	45
Insulation	35
Toxicity index (max.)	
Jacket	5
Fillers	5

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Insulation	1.5
Electrical moisture absorption	No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (minimum)
All sizes	100

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army – MI
Navy – SH

Preparing Activity:
Navy – SH
(Project 6145-2008-029)

Review Activities:
Army – AR, AV, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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